Before we dive into the mobile app development career path, it is very important to understand that this domain has changed a lot and there are several types of mobile apps. Developers can choose the kind of app they will work on, depending on their skillsets.

Types of mobile apps

Native apps: Earlier, developers designed native mobile apps, which are software programs that work on a particular platform or device. Native apps reside on the device and are accessed via icons on the home screen of a smart device. They can be installed via an app like Google Play or Apple App Store. Native apps provide optimised performance and take advantage of the latest technologies such as GPS, cameras, a compass, accelerometer, and so on. These apps can even use the device's notification system and work offline.

For iOS development, you require the following skills, tools and hardware:

- *Programming language:* Swift/Objective C
- *Tools:* XCODE
- *Hardware:* MacBook, iMac or MacPro

For Android development, you need the following:

- *Programming language:* Java/Kotlin
- *Tools:* Android Studio/Eclipse IDE
- *Hardware:* Any Windows, Linux or Mac compatible device

Native cross-platform apps: Native cross-platform apps are created when using APIs provided by Apple or Android SDK, but implementation is done in other programming languages not being supported by the operating system vendor. Usually, a third-party vendor provides an integrated development environment (IDE) that handles the process of creating the native application bundle for iOS and Android from a single cross-platform codebase.

To develop native cross-platform apps, you need the following:

- *Language:* JavaScript, Dart, .NET, C#
- *Library/framework:* React
- *Tools:* XCODE, Android Studio

Hybrid apps (HTML5 cross-platform apps): A hybrid app is a blend of both native and Web solutions. Unlike apps designed with React Native or Flutter, hybrid apps don't have access to the native features of the mobile device, out-of-the-box. Hybrid apps make it possible to embed HTML5 apps inside a thin native container, combining the best and worst elements of native and HTML5 apps. They use external plugins like Apache Cordova to integrate native features to the app. Hybrid apps render their graphic elements via the browser, which takes several steps to start showing the component on the screen. These are essentially Web views. With hybrid technologies like Ionic, NativeScript, Xaramin, React Native and others, it is easy to convert the Web app into a PWA that can be downloaded just like any other mobile app. All these options provide robust UI components that look and feel like their native counterparts, giving you a full suite of building blocks for your application.

What you need to develop hybrid apps is listed below.

- *Programming language:* HTML, JavaScript and CSS
- *Frameworks:* Angular, React, Vue
- *Tools:* Apache Cordova

Table 1 highlights the differences between native, cross-platform native and hybrid apps.

The mobile app development process

Developing mobile apps involves a six-stage process.

1. **Strategy:** The foundation for the app is proper strategy. In this phase, app developers have to identify the end users, research the competitors, and clearly define the app's goals and objectives. With a sound strategy, the vision of the app becomes clear and, after this, the next phase of the development process starts.
2. **Analysis and planning:** After developing a clear strategy, proper analysis and planning is required to ensure the app takes proper shape. This starts with defining use cases and capturing detailed functional requirements. In this stage, the developer has to prioritise the mobile app's requirements, and group these into delivery milestones before defining a minimum viable product (MVP). Proper

Table 1

Basis of difference	Native apps	Cross-platform native apps	Hybrid apps
Code base	Multiple code bases	Single code base	Single code base
App rendering engine	Native	Native	Browser
Performance	Best performance. Ideal for those apps that require heavy performance	Good, but not as good as single platform native app	Slow performance as compared to others
Cross-platform	No	Yes	Yes
Code re-use	No	Yes	Yes
Device access	Full	Limited	Full (with plugins)